Homework #6

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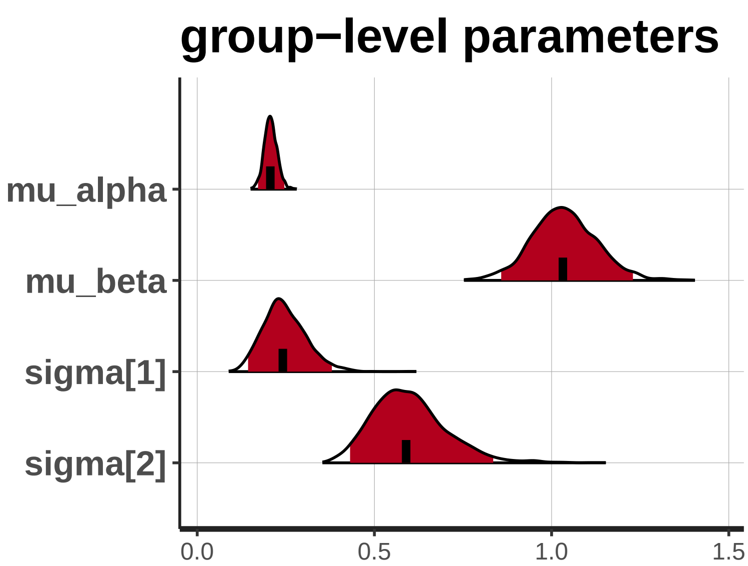
Q1.

(1)

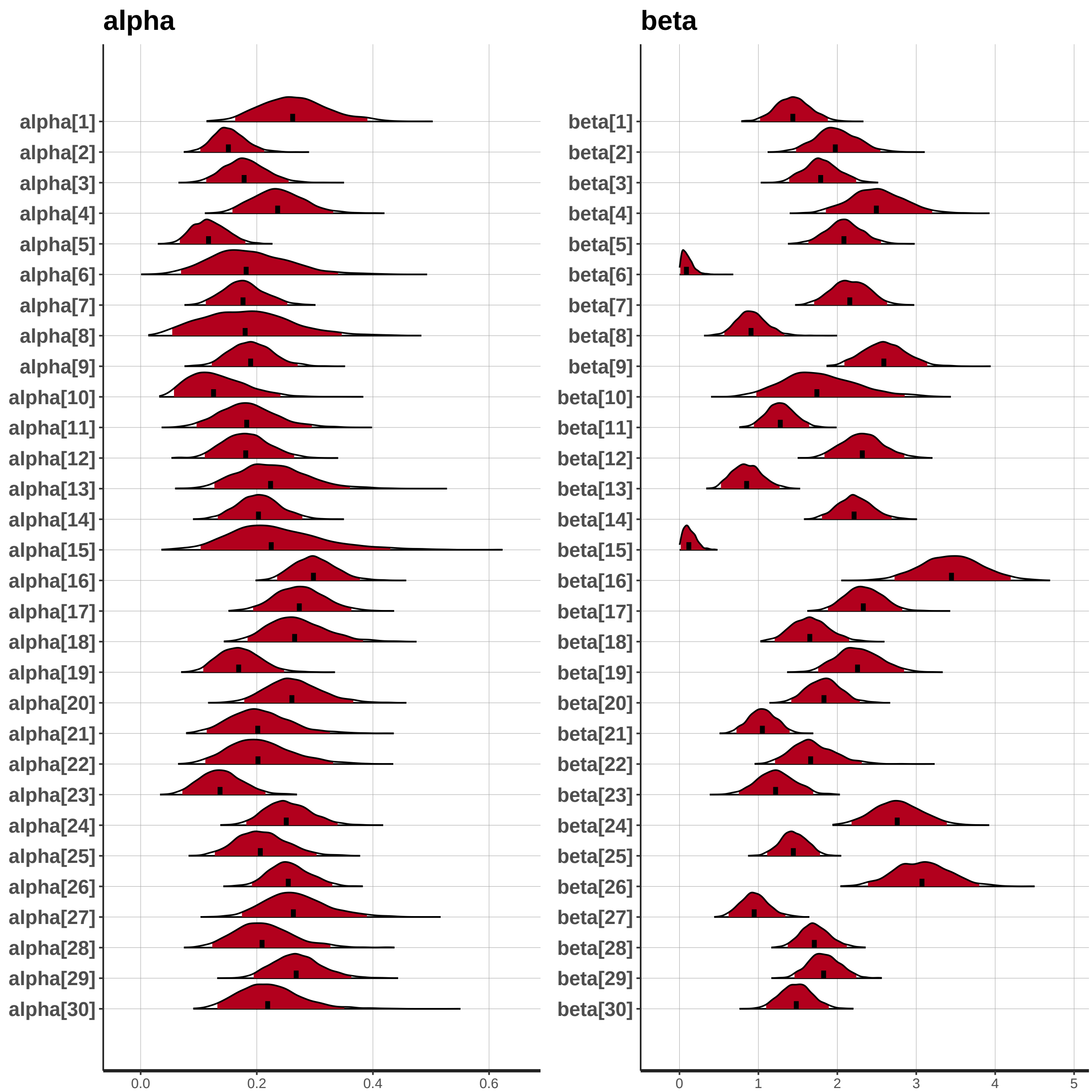
1. run `simulate_hw6_model1.R` and create `simul_data_hw6_model1.txt`
2. See codes in `Q1` folder

(2a)

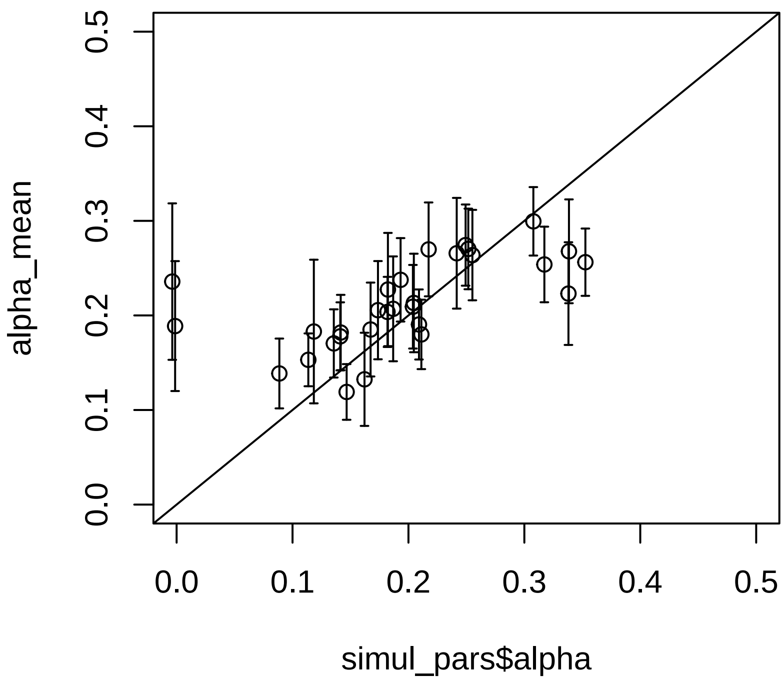
Posterior distributions of group-level parameters

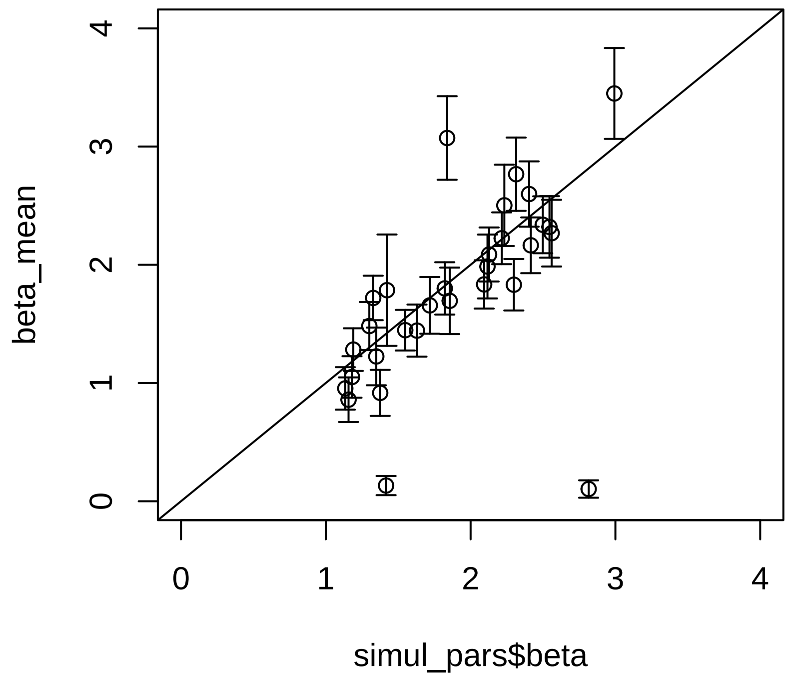


Posterior distributions of individual-level parameters



(2b) Scatterplot of true parameters and estimated parameters



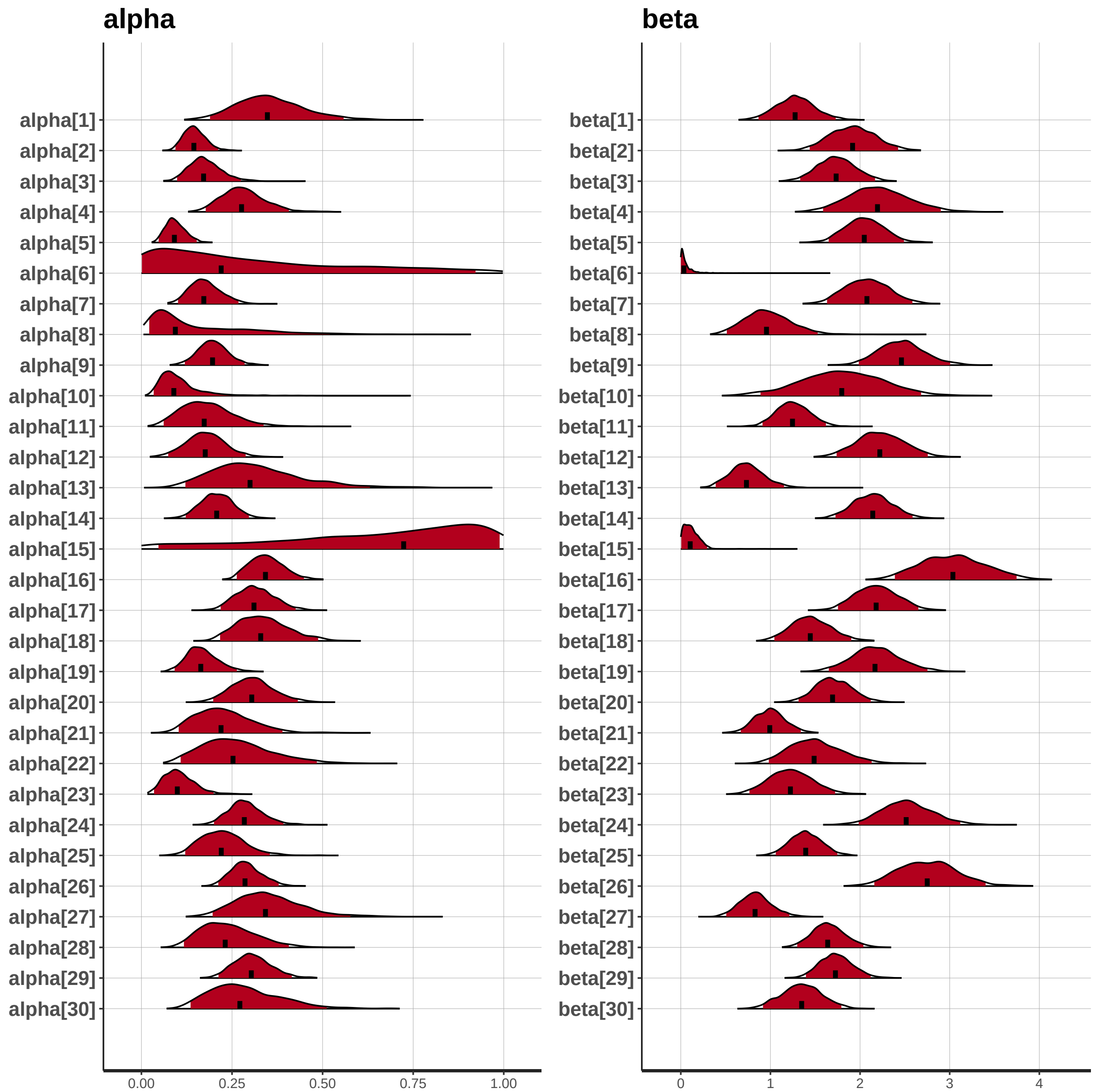


beta_mean
simul_pars$beta
0
1
2
3
4

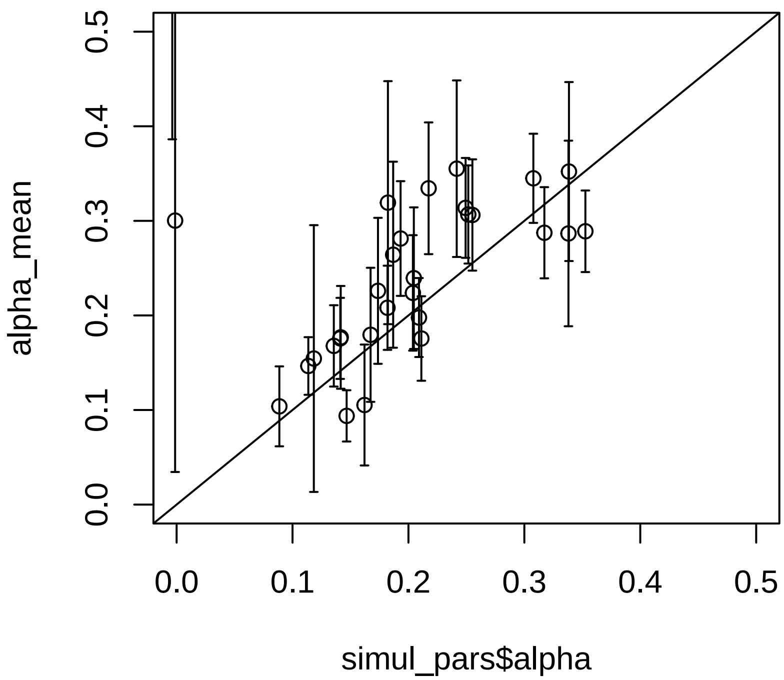
Q2

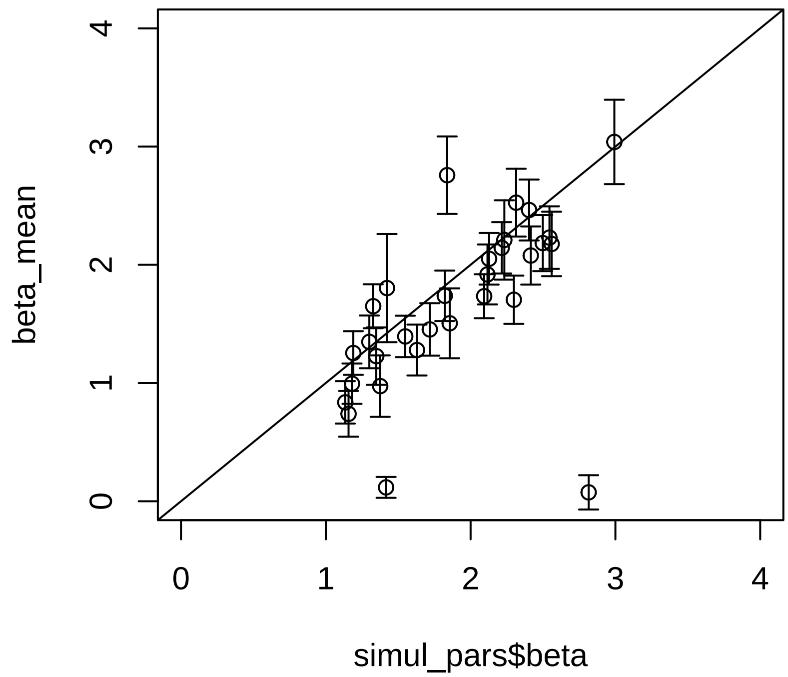
(1) Posterior distributions of the non-hierarchical model

see codes in `Q2` folder



(2) Scatterplot of true parameters and estimated parameters





(3) Noticeable difference between Q1 vs. Q2?

The scatterplots of the hierarchical model (Q1) are more concentrated and the results of parameter recovery are better than those of the non-hierarchicl model (Q2). This difference is due to the shrinkage

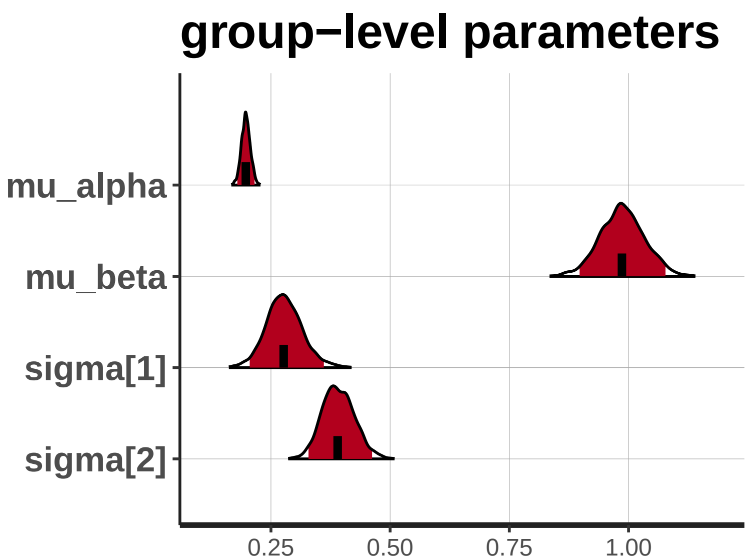
Q3

(1)

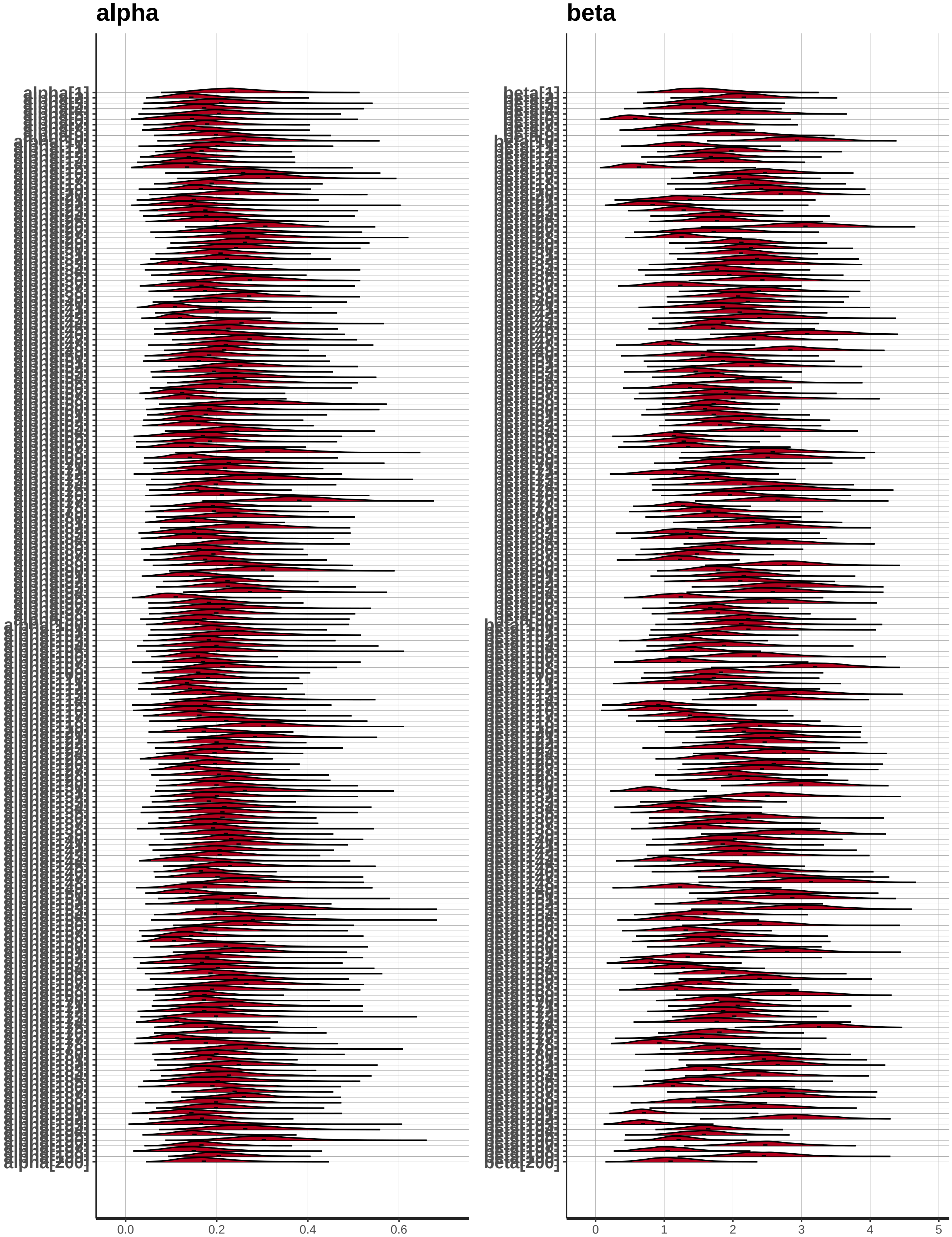
1. For simulation, see `simulate_hw7_model2.R`
2. For parameter estimation, see codes in `Q3` folder

(2) Posterior distributions

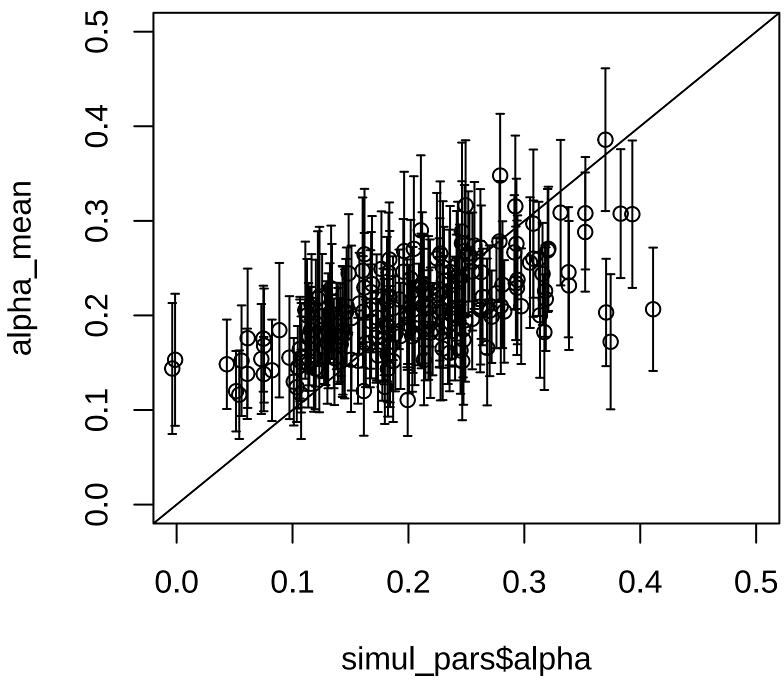
Posterior distributions of group-level parameters

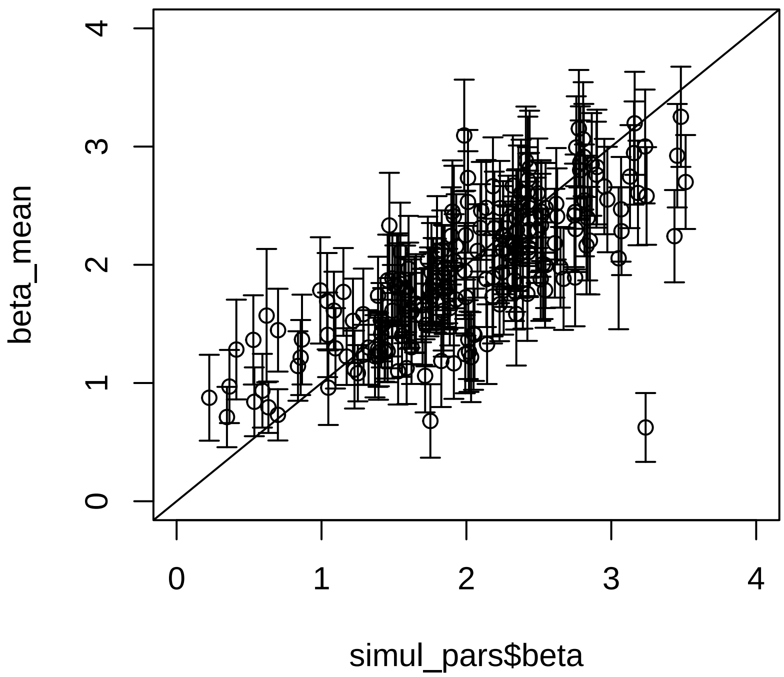


Posterior distributions of individual-level parameters



(3) Scatterplot of true parameters and estimated parameters





(4) Reason for the difference in shrinkage between parameters?

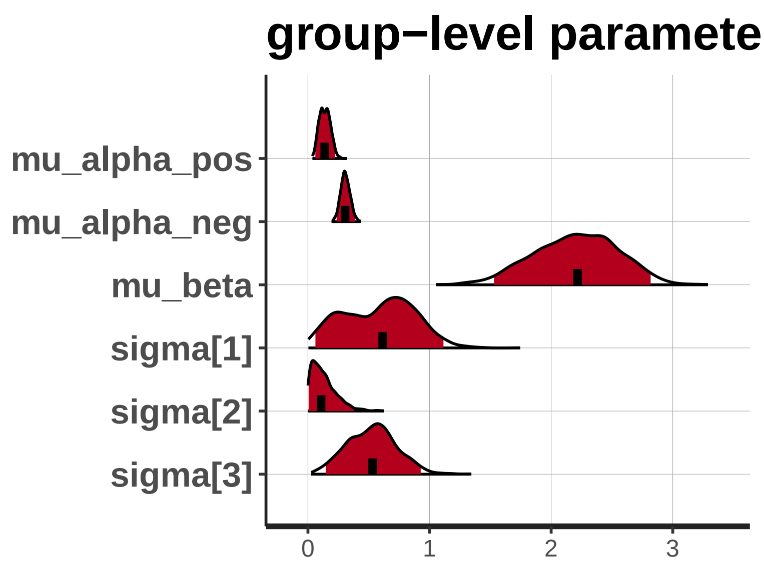
As can be seen in the scatterplots, the estimation of the individual parameter alpha seems to be more shrinked toward the value of mu_alpha than parameter beta. This is because the degree of shrinkage is greater for small coefficients compared to large coefficients.

Q4

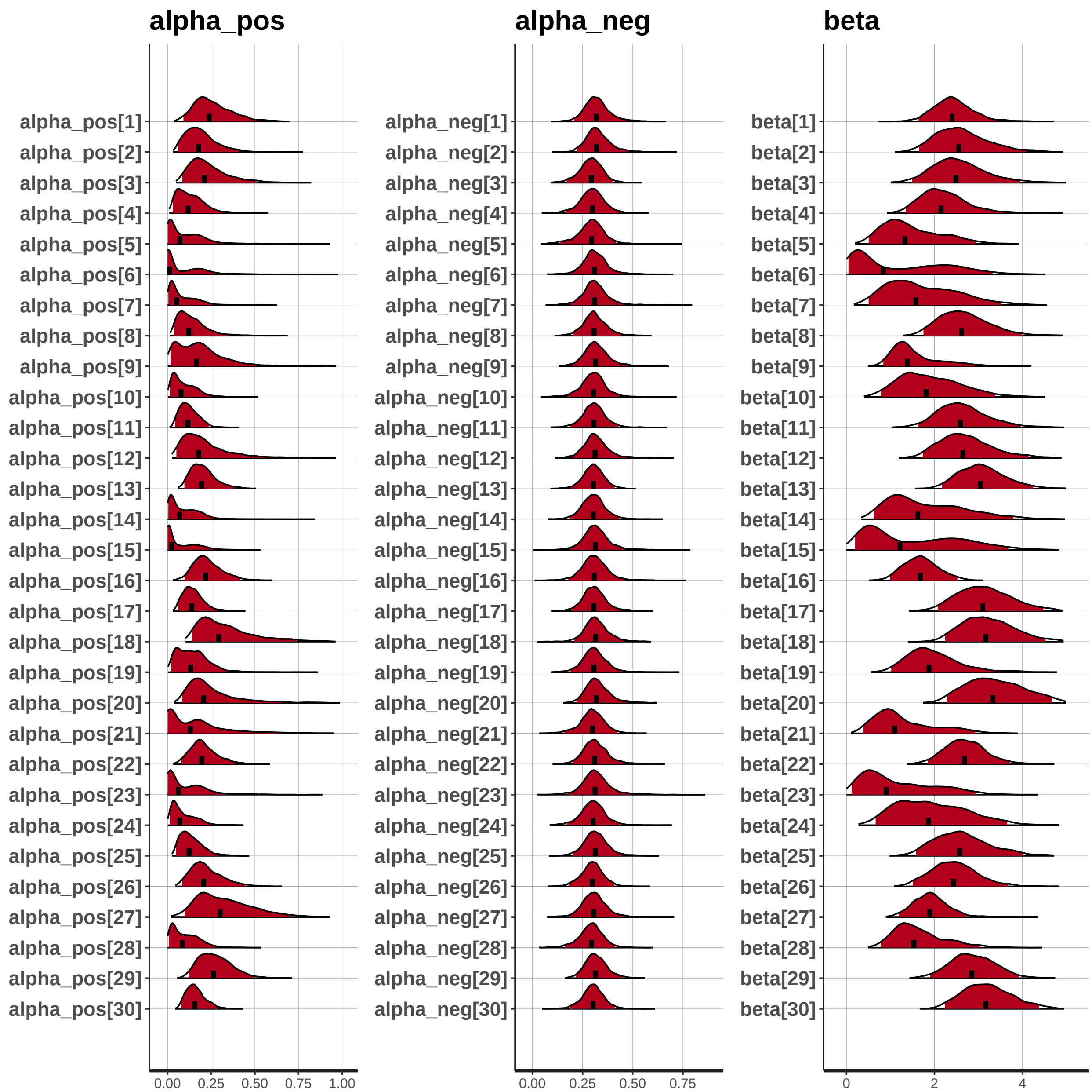
(1)

1. For simulation, `simulate_Q4.R`.
2. For parameter estimation, see `hw6_Q4.R` & `hw6_Q4.stan` in `Q4` folder.

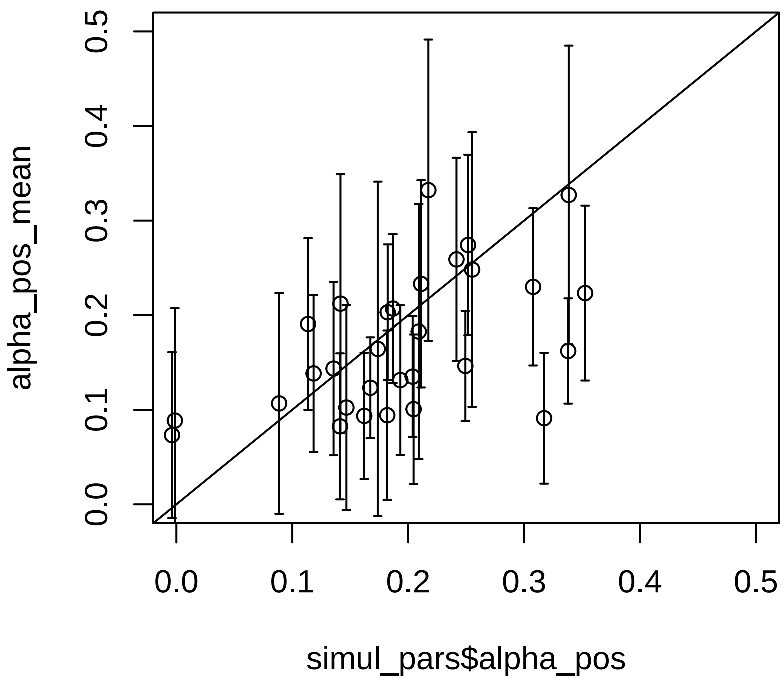
Posterior distributions of group-level parameters

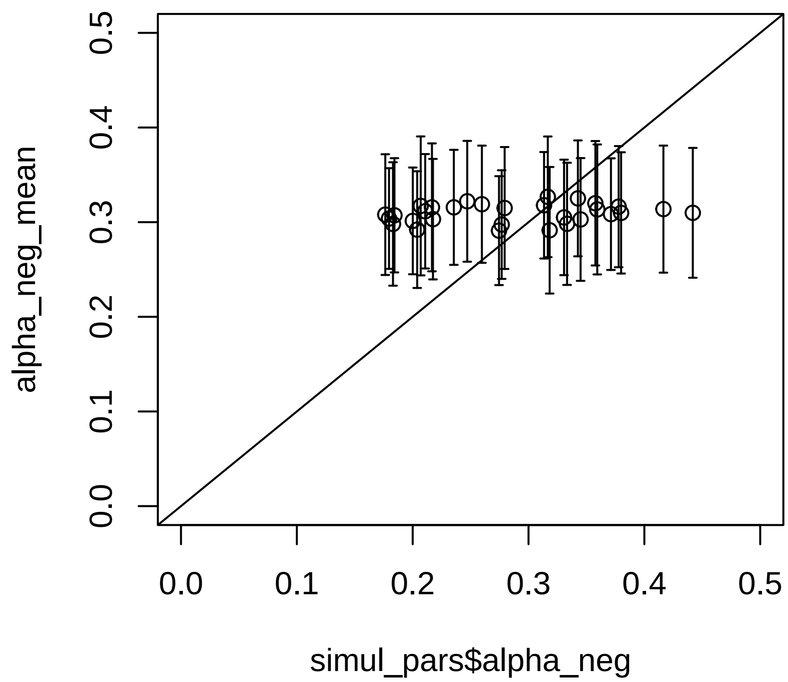


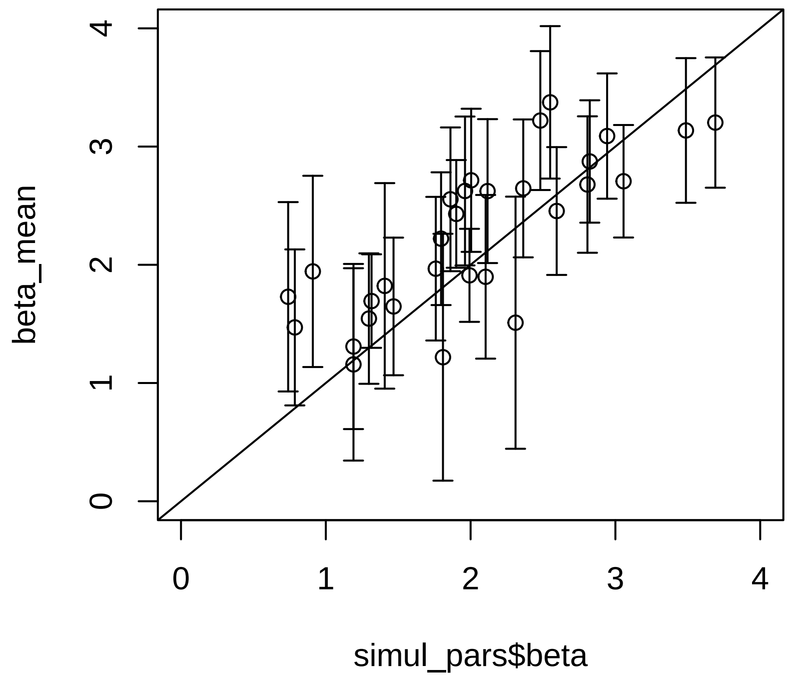
Posterior distributions of individual-level parameters



(2) Scatterplot of true parameters and estimated parameters



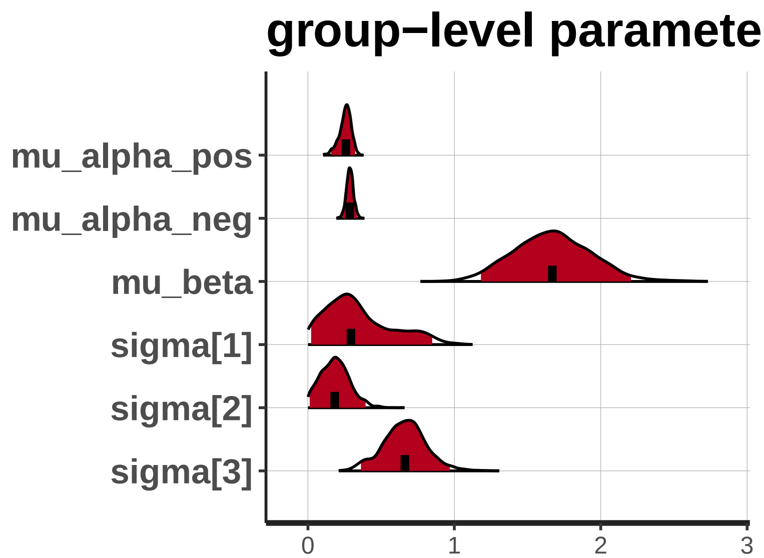

0.0
0.1
0.2
0.3
0.4
0.5
alpha_neg_mean
0.0
0.1
0.2
0.3
0.4
0.5
simul_pars$alpha_neg



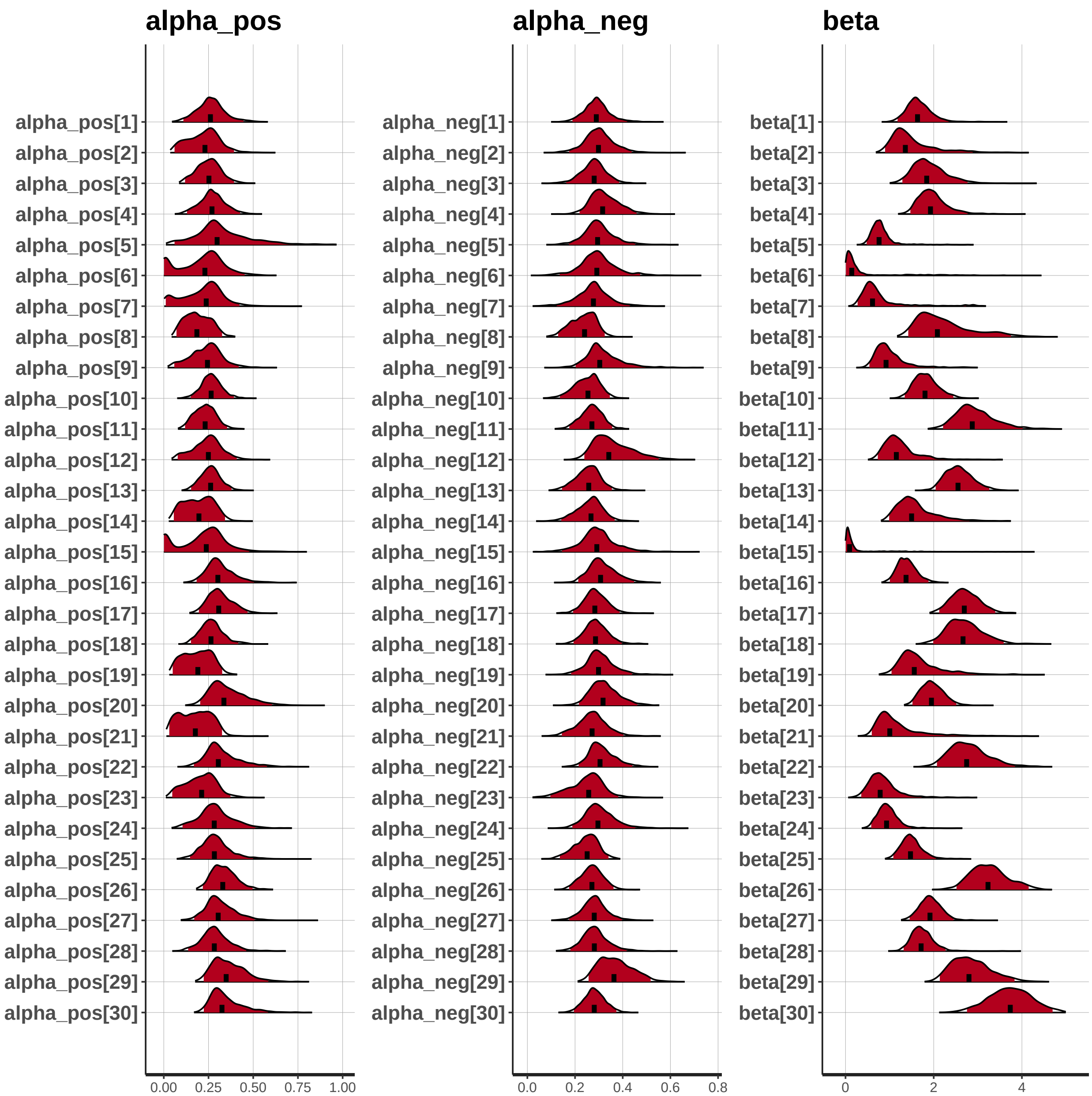
(3)

1. For simulation, see `simulate__hw7_model4_Q4_2.R`.
2. For parameter estimation, see `hw6_Q4_2.R` & `hw6_Q4.stan` in `Q4` folder.

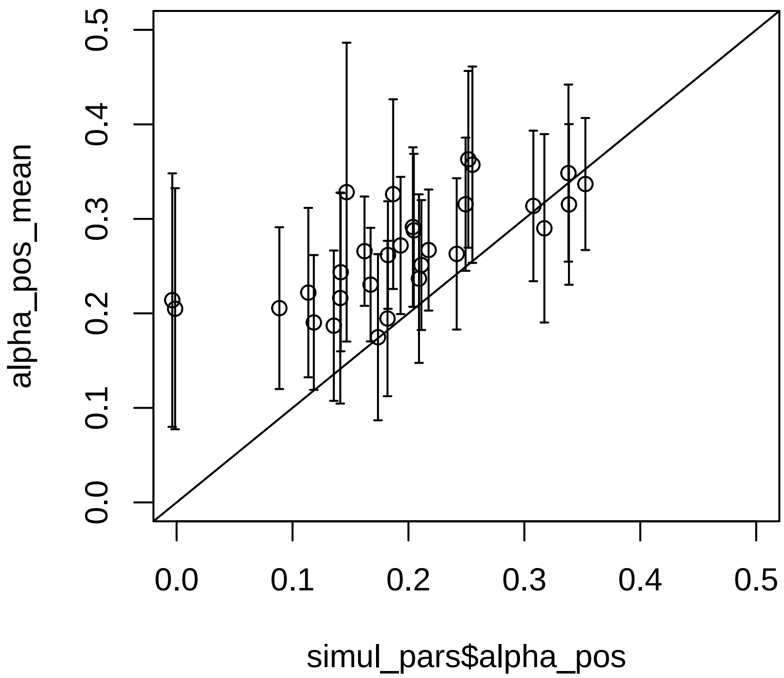
Posterior distributions of group-level parameters

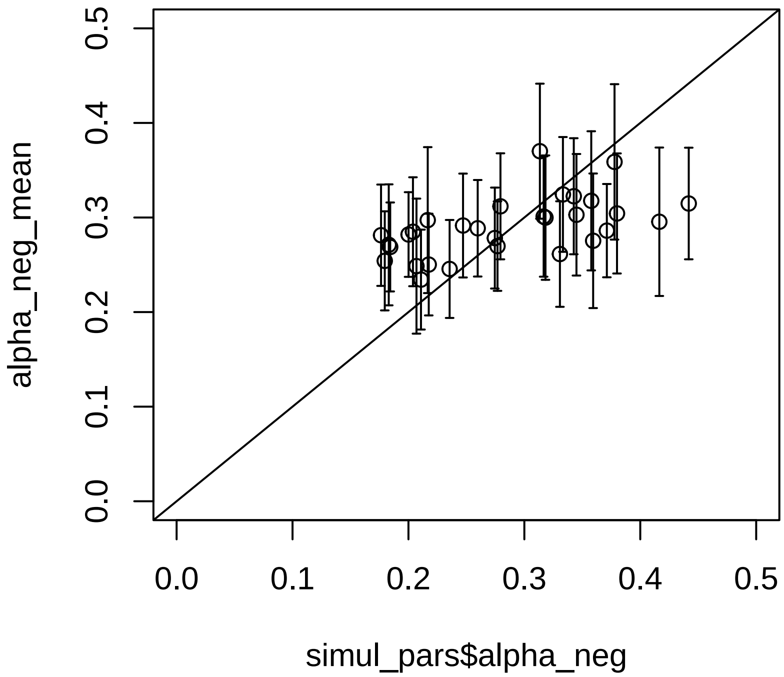


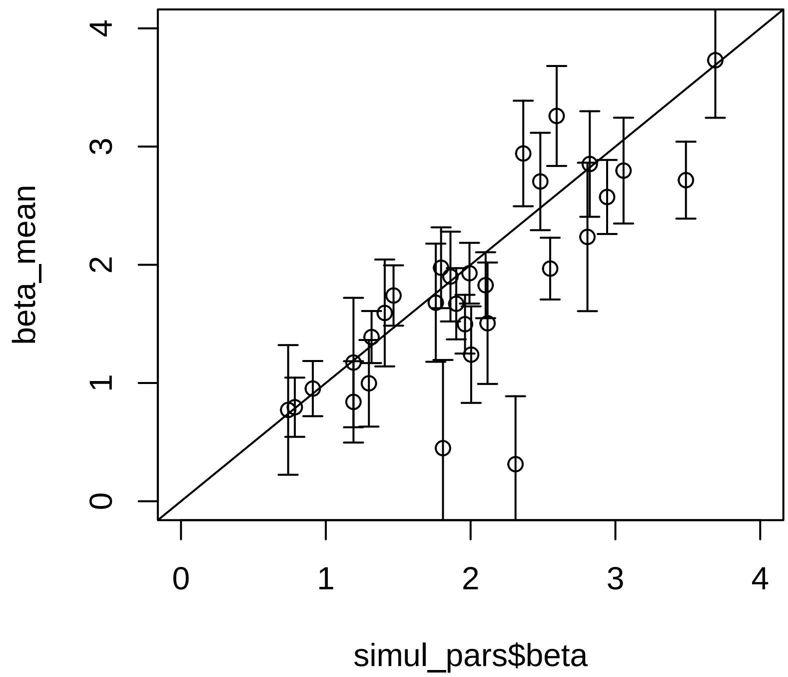
Posterior distributions of individual-level parameters



Scatterplot of true parameters and estimated parameters




alpha_neg_mean
0.0
0.1
0.2
0.3
0.4
0.5
0.0
0.1
0.2
0.3
0.4
0.5
simul_pars$alpha_neg



(4) Difference between Q4(2) vs. Q4(3)?

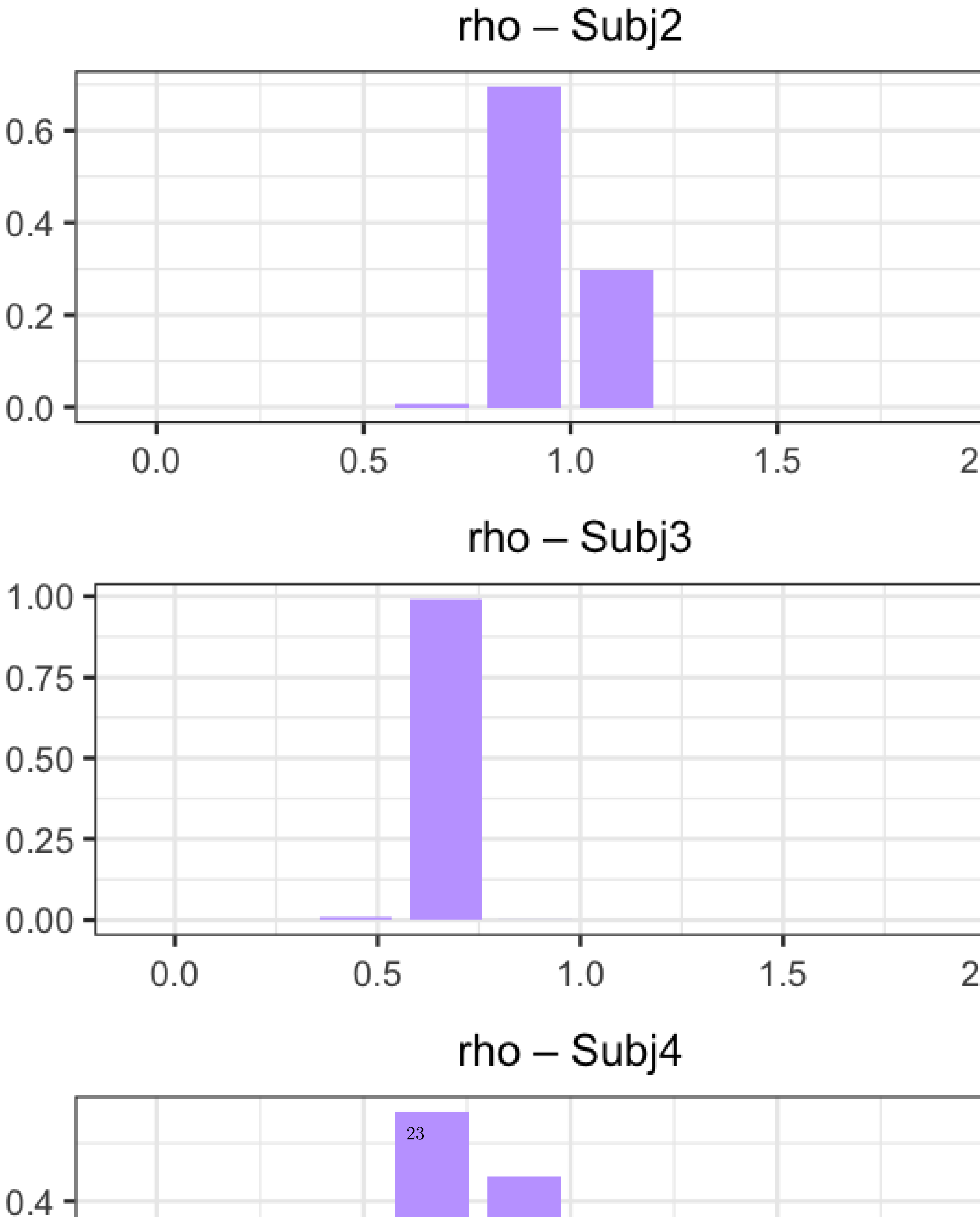
Q4(3) showed better parameter recovery results than Q4(2), for the alpha parameters. One reason could be the increase in the number of trials in Q4(3), which increases the information for each individual. The low estimation accuracy of `alpha_neg` is a consequence of setting relatively high `pr_correct` in Q4(2) compared to Q4(3). Thus, there was relatively smaller information for negative PE in Q4(2) while positive and negative PE were fairly well distributed in Q4(3).

Q3

(1) Grid approximation and posterior distributions of the ra_prospect model - 10 Grids

Table 1: Posterior means (10-grid)

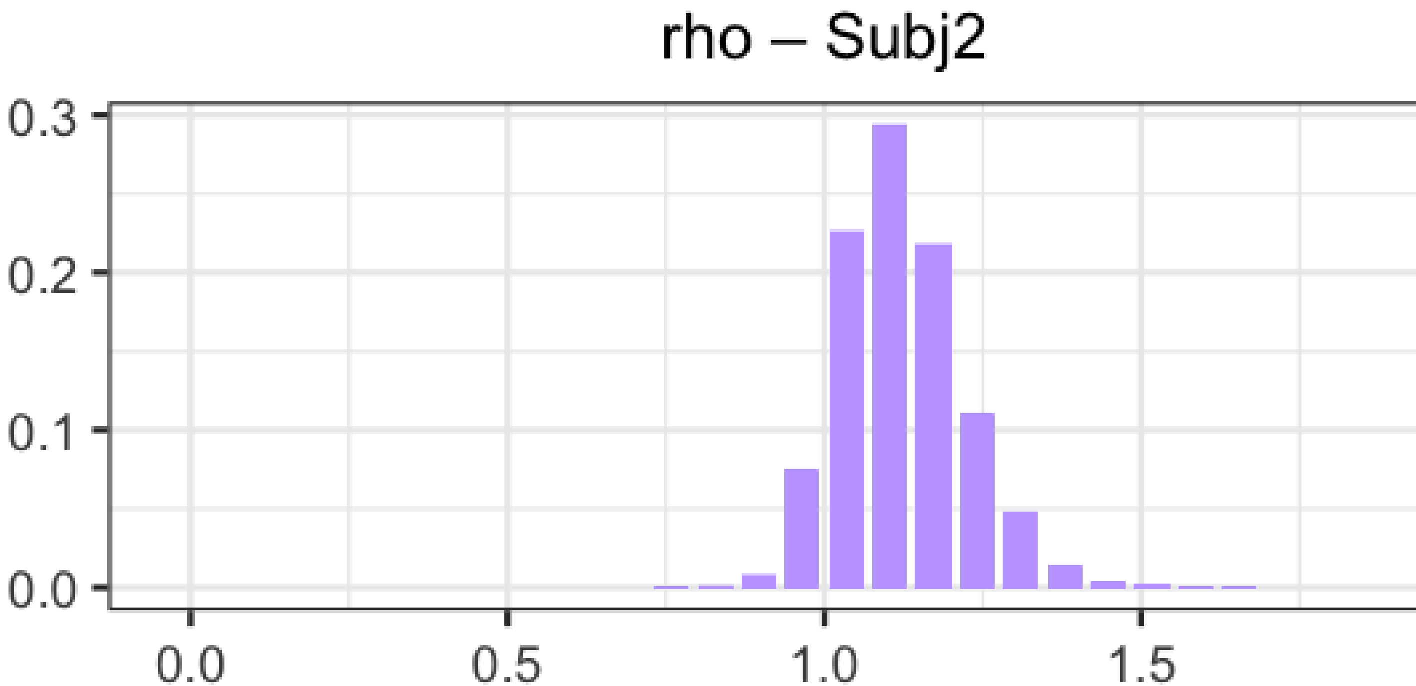
subjID	rho	lambda	tau
2	0.954	1.111	0.669
3	0.665	2.263	4.215
4	0.765	1.111	1.060
6	0.889	1.111	2.322
7	0.958	1.341	1.940

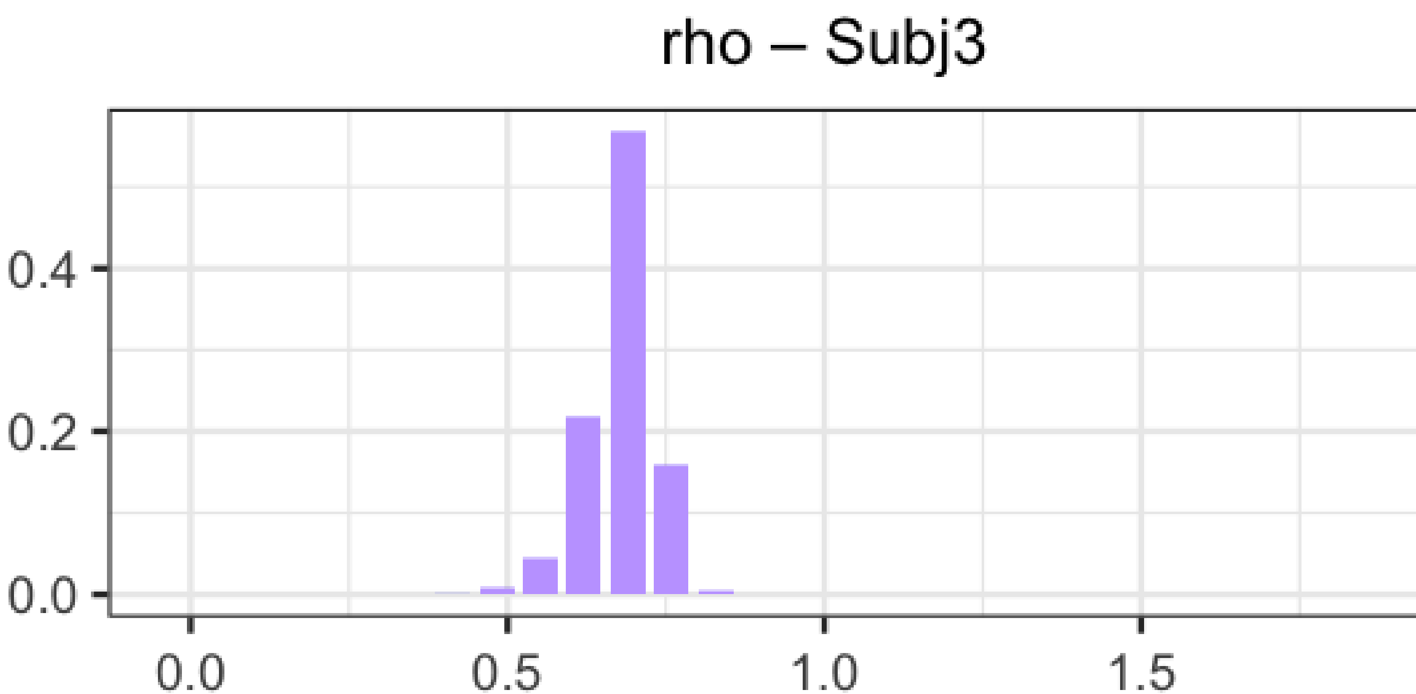

rho – Subj2
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
rho – Subj3
1.00
0.75
0.50
0.25
0.00
0.0
0.5
1.0
1.5
rho – Subj4
0.4

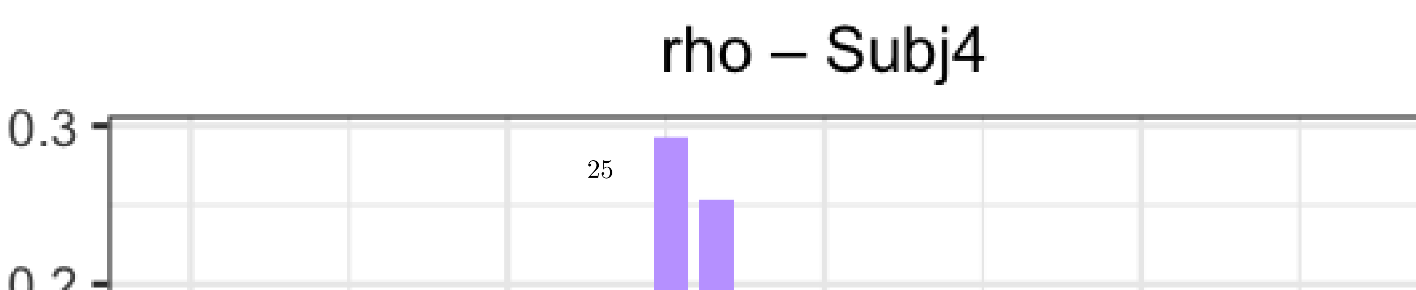
(2) Grid approximation and posterior distributions of the ra_prospect model - 30 Grids

Table 2: Posterior means (30-grid)

subjID	rho	lambda	tau
2	1.122	0.691	0.760
3	0.678	2.570	3.740
4	0.791	1.073	1.094
6	0.872	1.034	2.900
7	0.962	1.389	2.486


rho – Subj2
0.3
0.2
0.1
0.0
0.0
0.5
1.0
1.5


rho – Subj3
0.4
0.2
0.0
0.0
0.5
1.0
1.5


rho – Subj4
0.3